

1. Find two angles co-terminal with $\frac{4\pi}{3}$. .../1-K Answers may vary $\theta_1 = \frac{4\pi}{3} + 2\pi = \frac{10\pi}{3}$ $\theta_2 = \frac{4\pi}{3} - 2\pi = -\frac{2\pi}{3}$	2. Find the exact value for the $2\sin^2\left(\frac{3\pi}{8}\right) - 1$: .../2-K $2\sin^2\left(\frac{3\pi}{8}\right) - 1 = -\left[1 - 2\sin^2\left(\frac{3\pi}{8}\right)\right] = -\cos\left[\pi - \frac{\pi}{4}\right]$ $= -\cos\left[2\left(\frac{3\pi}{8}\right)\right] = -\cos\left(\frac{3\pi}{4}\right) = -(-\cos\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$
3. Convert $\frac{5\pi}{24}$ to degrees .../1-K $\frac{5\pi}{24} \cdot \frac{180^\circ}{\pi} = 37.5^\circ$	4. Convert -55° to radians .../1-K $-55^\circ \left(\frac{\pi}{180^\circ}\right) = -\frac{11\pi}{36}$
Multiple Choice: Circle one answer that best answers each question #5-6	
5. A circle has a radius of 20 cm . The exact length of arc that subtends by a central angle of 135° is:.../1-K $\hookrightarrow 135\left(\frac{\pi}{180}\right) = \frac{3\pi}{4}$ rads a) 2700 cm b) 50 cm c) 15 cm d) <u>15πcm</u> $s = 20\left(\frac{3\pi}{4}\right) = 15\pi$	6. $2\sin\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{4}\right)$ is equivalent to: .../1-K a) <u>$\sin\left(\frac{\pi}{2}\right)$</u> $\hookrightarrow \sin\left[2\left(\frac{\pi}{4}\right)\right] = \sin\left(\frac{\pi}{2}\right)$ b) $\cos\left(\frac{\pi}{2}\right)$ c) $\sin\left(\frac{\pi}{8}\right)$ d) $\cos\left(\frac{\pi}{8}\right)$
7. Completely simplify $\frac{\sin\left(\frac{\pi}{2} - x\right)\cot\left(\frac{\pi}{2} + x\right)}{\cos(x + \pi)} - \frac{\csc\left(\frac{3\pi}{2} - x\right)}{\sec(2\pi - x)}$. Show all steps. .../4-K $= \frac{\cancel{\cos(x)} \cdot [-\cancel{\tan(x)}]}{-\cancel{\cos(x)}} - \frac{-\cancel{\sec(x)}}{\cancel{\sec(x)}}$ $= \tan(x) + 1$	
8. Fin the exact value for the following. Show all steps. .../4-A $\sin\left(\frac{3\pi}{8}\right)$ $= \sin\left(\frac{\pi}{2} - \frac{\pi}{8}\right)$ $= \cos\left(\frac{\pi}{8}\right)$ $\cos\left(\frac{\pi}{4}\right) = 2\cos^2\left(\frac{\pi}{8}\right) - 1$ $\frac{\sqrt{2}}{2} = 2\cos^2\left(\frac{\pi}{8}\right) - 1$ $\frac{\sqrt{2}}{2} + 1 = 2\cos^2\left(\frac{\pi}{8}\right)$ $\frac{\sqrt{2} + 2}{2} = 2\cos^2\left(\frac{\pi}{8}\right)$ $\sqrt{\frac{\sqrt{2} + 2}{4}} = \sqrt{\cos^2\left(\frac{\pi}{8}\right)}$ $\therefore \frac{\sqrt{\sqrt{2} + 2}}{2} = \cos\left(\frac{\pi}{8}\right)$	$\sin\left(\frac{7\pi}{12}\right)$.../3-A $= \sin\left(\frac{\pi}{3} + \frac{\pi}{4}\right)$ $= \sin\frac{\pi}{3}\cos\frac{\pi}{4} + \cos\frac{\pi}{3}\sin\frac{\pi}{4}$ $= \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$ $= \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4}$ $= \frac{\sqrt{6} + \sqrt{2}}{4}$
9. Prove $\frac{\sin(2x)}{1 + \cos(2x)} = \frac{\tan(x)\sin(2x)}{2\sin(x)\cos(x)}$./4-A LS = $\frac{2\sin(x)\cos(x)}{1 + [2\cos^2(x) - 1]}$ $= \frac{\cancel{2}\sin(x)\cancel{\cos(x)}}{\cancel{2}\cos^2(x)}$ $= \frac{\sin(x)}{\cos(x)}$ $= \tan(x) \cdot \frac{\sin(2x)}{\sin(2x)}$ $= \frac{\tan(x)\sin(2x)}{\sin(2x)}$ $= \frac{\tan(x)\sin(2x)}{2\sin(x)\cos(x)}$ = RS	

10. Determine the **exact** value of $\frac{\csc\left(\frac{5\pi}{3}\right) + \tan\left(\frac{\pi}{6}\right)}{\sin\left(\frac{5\pi}{4}\right) \cot\left(\frac{\pi}{3}\right)}$ (rationalize if necessary). .../4-A

$$= \frac{\csc\left(2\pi - \frac{\pi}{3}\right) + \frac{1}{\sqrt{3}}}{\sin\left(\pi + \frac{\pi}{4}\right) \cdot \frac{1}{\sqrt{3}}}$$
$$= \frac{-\csc\left(\frac{\pi}{3}\right) + \frac{\sqrt{3}}{3}}{-\sin\left(\frac{\pi}{4}\right) \cdot \frac{1}{\sqrt{3}}}$$
$$= \frac{-\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}}{-\frac{\frac{\sqrt{2}}{2}}{1} \cdot \frac{1}{\sqrt{3}}}$$

$$= \frac{-\frac{1}{\sqrt{3}}}{-\frac{\sqrt{2}}{2\sqrt{3}}}$$
$$= \frac{1}{\cancel{\sqrt{3}}} \cdot \frac{2\sqrt{3}}{\sqrt{2}}$$
$$= \frac{2}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}}$$
$$= \sqrt{2}$$

11. If $\tan(x) = -\frac{3}{4}$, and $\frac{3\pi}{2} < x < 2\pi$, then determine the value of $\tan\left(\frac{x}{2}\right)$/5-T

$$\tan(x) = \frac{2\tan\left(\frac{x}{2}\right)}{1 - \tan^2\left(\frac{x}{2}\right)}$$
$$-\frac{3}{4} = \frac{2\tan\left(\frac{x}{2}\right)}{1 - \tan^2\left(\frac{x}{2}\right)}$$

Let $A = \tan\left(\frac{x}{2}\right)$:

$$-\frac{3}{4} = \frac{2A}{1 - A^2}$$
$$-3(1 - A^2) = 8A$$
$$-3 + 3A^2 = 8A$$
$$3A^2 - 8A - 3 = 0$$
$$(3A + 1)(A - 3) = 0$$

$$3A + 1 = 0$$
$$A = -\frac{1}{3}$$
$$\tan\left(\frac{x}{2}\right) = -\frac{1}{3}$$

$$A - 3 = 0$$
$$A = 3$$
$$\tan\left(\frac{x}{2}\right) = 3$$

since the tangent is negative in Q4
↳ inadmissible since $\frac{3\pi}{2} < x < 2\pi$

$$\therefore \tan\left(\frac{x}{2}\right) = -\frac{1}{3}$$

12. If $\cos(\theta) + \sin(\theta) = \frac{1 + \sqrt{3}}{2}$ — ① and $\cos(\theta) - \sin(\theta) = \frac{1 - \sqrt{3}}{2}$ — ② find the value of $\sin(2\theta)$/5-T

$$\textcircled{1} + \textcircled{2}: \cos\theta + \sin\theta = \frac{1 + \sqrt{3}}{2}$$
$$+ \textcircled{2} \cos\theta - \sin\theta = \frac{1 - \sqrt{3}}{2}$$
$$2\cos\theta = \frac{2}{2}$$
$$\cos\theta = \frac{1}{2}$$

Sub. $\cos\theta = \frac{1}{2}$ into ①: $\frac{1}{2} + \sin(\theta) = \frac{1 + \sqrt{3}}{2}$

$$\sin(\theta) = \frac{\sqrt{3}}{2}$$

$$\sin(2\theta) = 2\sin\theta \cos\theta$$
$$= 2\left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{2}\right)$$
$$= \frac{\sqrt{3}}{2}$$
$$\therefore \sin(2\theta) = \frac{\sqrt{3}}{2}$$